

Viva Preparation Guide

Prompt Engineering for Generative AI — Case Study 1 & Case Study 2

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0. Do These Things BEFORE the Viva (Not Optional)

Both documents contain explicit notes that they are working drafts. If an examiner probes and you can't back these up, it will show immediately.

Document	What it admits is still incomplete	What you must do
Case Study 1	Phase 9 (Multi-LLM Comparison) says "[Record after running Version 4 prompt]" — it was never actually run	Actually run the Phase 8 Version 4 prompt on ChatGPT and Claude yourself. Screenshot both outputs. Fill in real scores.
Case Study 1	AI Response reliability scores (2, 3, 4) are described as "generated for this case study"	Be ready to show the actual prompt/response pairs if asked — even a screenshot on your phone helps
Case Study 2	"How to Use This Document" explicitly says citation counts are qualitative estimates, not live counts, and must be "pulled live at submission time"	Pull real citation counts for at least the 5–6 papers you'd discuss most (P1 Lewis et al., P13 GraphRAG, P11 Self-RAG, P16 RAPTOR) from Google Scholar
Case Study 2	It says the review "synthesises each paper's abstract... a genuine literature review still requires reading the full papers"	Actually read the abstracts + intros of your 5–6 most-citable papers in full. Examiners love asking about the <i>method</i> section, not just the abstract line you copied into the matrix

If you skip this, the safest honest answer to "did you run this yourself?" is: *"I used AI to help draft the structure and I've since verified the sources; some of the live-tool comparisons I'm finalizing"* — not a denial, not a bluff.

1. Core Concepts You Must Be Able to Explain Cold

These will almost certainly be asked regardless of which case study comes up, because they're the actual subject of the course.

Term	One-line definition	Give this example if asked
Prompt Engineering	Deliberately designing/refining an input to an LLM to reliably get accurate, well-scoped output	Zero-shot → Role → CoT → ReAct progression in Phase 8 of CS1
Zero-shot prompting	Asking the question directly, no examples, no persona, no structure	"How many jobs has AI caused to be lost in 2026?"
Role/Persona prompting	Instructing the model to act as a specific expert to constrain tone, rigor, and domain framing	"Act as a Senior Investigative Journalist..." (CS1 Phase 4)
Chain-of-Thought (CoT)	Instructing the model to reason step-by-step <i>before</i> concluding, exposing intermediate logic	"Think step by step. For each source, extract the claim, time period, type of data..." (CS1 Phase 5)
ReAct (Reason + Act)	A loop of Thought → Action (retrieve/verify) → Observation → repeat, used for claim-by-claim verification	Phase 6 of CS1 — reasoning about what evidence would confirm a claim, then checking sources
Prompt Chaining	Breaking one complex task into a sequence of smaller prompts, each output feeding the next	CS1 Phase 10 (Topic → Extraction → Claims → Contradictions → Verification → Bias → Report)
Hallucination	A fluent, plausible-sounding claim that is not actually supported by any real source	CS1 Phase 7: "DOL confirms 85 million AI job losses" — invented
Sensationalism/Recency bias	Model or aggregator defaults to the largest/most dramatic or most recently-labeled number, regardless of accuracy	CS1: 2023's 85M figure re-dated to 2026 by an aggregator
RAG (Retrieval-Augmented Generation)	Grounding an LLM's output in passages retrieved from an external corpus at inference time, instead of relying only on parametric memory	CS2 P1, Lewis et al. 2020
Hybrid retrieval	Combining lexical/sparse (BM25) and dense/semantic (embedding) retrieval, often with a reranker	CS2 P27 Re2G — the direct precedent for "hybrid RAG"
GraphRAG	Building an entity/knowledge graph + hierarchical summaries from a corpus so an	CS2 P13, Microsoft

Term	One-line definition	Give this example if asked
	LLM can answer "what are the themes"	questions flat retrieval can't
If you remember nothing else: be able to explain <i>why</i> each technique reduces a <i>specific failure mode</i> (not just "it makes it better"). Zero-shot → dramatic-number bias. Role → source discipline. CoT → forces actual-vs-projected classification. ReAct → per-claim verification with confidence scores.		
2. Case Study 1 — "Are AI Job Losses as Severe as Headlines Claim?"		
Facts you must have memorized		
Topic category: Technology & AI, secondary Business & Economy		
Three key sources and what they actually measure (this is the whole point of the case study — examiners will test whether you understand <i>why they're not contradictory</i>):		
WEF Future of Jobs Report 2025: 92 million jobs displaced, 170 million created by 2030 (net +78M) — a <i>decade-long structural projection</i>		
Goldman Sachs: ~16,000 net US jobs/month lost to AI as of April 2026 — a <i>private estimate</i>, not government-verified		
Challenger, Gray & Christmas: ~54,694–55,000 US layoffs <i>directly attributed</i> to AI in 2025, out of 800,000+ total announced cuts — AI is <i>one of several</i> causes, not the dominant one		
The core misinformation mechanism found: a 2023 WEF projection (85 million jobs displaced <i>by 2025</i>) recycled by aggregator sites with the date simply pushed to 2026 — an old forecast kept in circulation past its useful life, not a fresh finding		
Hallucination audit result: 15 claims tested → 9 verified, 2 low-confidence/unverified, 4 hallucinated (fake DOL confirmation, 100% customer-service-job claim, fake G20 legislation, fake central-bank inflation-forecast claim)		
Score progression across prompt versions: Accuracy 2→3→4→4, Reliability 2→3→4→5, Bias Reduction 1→2→3→4		
Likely viva questions		
"Why isn't the WEF's 92 million figure a contradiction of Goldman Sachs's 16,000/month?" → Different timeframe (10-year projection vs. current monthly estimate) and different scope (global structural trend vs. US-specific current displacement). This is the central insight of the whole case study — you must be able to say this fluently.		
"What does BLS actually track, and why does that matter here?" → The U.S. Bureau of Labor Statistics has no distinct category for "AI-caused" layoffs, which is exactly why private estimates of uneven quality fill the gap.		
"Give me an example of a hallucination you found and explain why it's dangerous, not just wrong." → E.g. the fake DOL attribution — dangerous because it borrows the authority of an official government source to make an unverified number look confirmed.		

"Why did role prompting alone not fully fix the bias?" → It forced source attribution and period-matching but still defaulted to citing the most alarming figure *first* before counter-evidence — bias control improved (2→3) but wasn't perfect until CoT + ReAct were layered in.

"What's the difference between an 'unverified' claim and a 'hallucinated' claim in your audit?" → Unverified = plausible, traces to a real (if uncertain) source, e.g. the 23-30% company-survey figure. Hallucinated = no supporting evidence exists at all, contradicted by other sources, or attributed to a body that never said it.

"If I ask you right now to verify one of these figures live, can you?" → Be ready to actually pull up the WEF Future of Jobs Report 2025 or a recent Challenger Gray & Christmas report if you have a laptop/phone — this is your strongest, most concrete source.

3. Case Study 2 — Hybrid RAG for Enterprise Knowledge Systems

Facts you must have memorized

Topic narrowing path: AI → Generative AI → RAG → Hybrid RAG → Enterprise RAG Systems

Foundational paper: Lewis et al. 2020, "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks" (NeurIPS) — joint DPR retriever + BART generator; established non-parametric memory as an alternative to ever-larger models

The "hybrid" precedent: Re2G (Glass et al. 2022) — fuses sparse (BM25) + dense retrieval, then reranks; directly named as the blueprint for what's now called hybrid RAG

Graph-based family: Microsoft GraphRAG (P13) builds entity graphs + community summaries for "what are the themes" queries; LightRAG (P15) cuts GraphRAG's indexing cost; RAPTOR (P16) recursively clusters/summarizes into a retrieval tree for long-document multi-hop QA; HippoRAG (P26) uses PageRank-style graph retrieval inspired by hippocampal memory

Self-verifying retrieval: Self-RAG (P11) trains the model to decide *when* to retrieve and critique its own output; CRAG (P12) adds a lightweight evaluator that grades retrieval confidence and falls back to web search

The two real enterprise case studies: Packowski et al. (IBM, P17) and Fatehkia et al. (T-RAG, P20) — both converge on the same finding: **chunking/content design matters more than which retriever or LLM you use**

Evaluation frameworks: RAGAs (P18, reference-free LLM-judged metrics), RGB benchmark (P29, stress-tests robustness to noise/contradiction), TREC RAG Track (P30, community-wide standardized benchmark)

Four research gaps (G1-G4):

G1: Enterprise-scale evaluation is thin (only P17, P20 are real deployments)

G2: Standard retrieval metrics (nDCG) correlate weakly with downstream answer accuracy

G3: Graph/tree indexes (GraphRAG, RAPTOR, HippoRAG) lack incremental update strategies — rebuilt in batch, go stale

G4: Robustness to noisy/contradictory enterprise documents is only tested on synthetic benchmarks

Three proposed novel ideas: (1) hybrid sparse+dense+graph router that picks strategy per query type, (2) incremental priority-queued index-refresh scheduler, (3) contradiction-aware retrieval evaluator for superseded/conflicting documents

Likely viva questions

"Explain BM25 in one sentence, and why does it still matter in 2026?" → A probabilistic term-weighting/lexical ranking function; it remains a zero-training-cost, exact-keyword-matching baseline that hybrid systems still fuse with dense retrieval because it catches things embeddings miss (product codes, exact clause numbers).

"What's the actual difference between GraphRAG and RAPTOR?" → GraphRAG builds an entity knowledge graph with community summaries (good for global/thematic queries across a corpus); RAPTOR recursively clusters and summarizes chunks into a tree (good for multi-hop reasoning within long documents). Different structures for related but distinct problems.

"Why do you say retrieval metrics don't predict downstream accuracy — what's the evidence?" → Salemi & Zamani (P24) directly show retrieval-stage nDCG correlates imperfectly with end-task QA accuracy — meaning a system can look good on retrieval metrics but still give worse answers.

"Your own industry papers (P17, P20) found chunking design matters more than model choice — doesn't that undercut the whole point of studying retrieval architectures?" → This is a fair challenge — be ready to say: it doesn't undercut the field, it reframes it. It means architecture choice sets the *ceiling* of what's possible, but content/chunking design determines whether you actually *reach* that ceiling in production — which is exactly why G1 (thin enterprise evaluation) matters.

"How did you avoid citation hallucination when using AI to find these 30 papers?" → Per Phase 4's activity log: AI was used to *locate* candidates and pull bibliographic details from the paper's own arXiv/DOI page or citation lists — never asked to recall citations from memory. Every entry was cross-checked against its own landing page.

"Pull up paper P1 and tell me what dataset section 4 of the actual paper uses, not just what's in your table." → This is where you need to have actually skimmed at least the top 5–6 papers beyond the abstract-level summary in the matrix.

4. Cross-Cutting Questions That Could Apply to Either Case Study

"What's the single biggest lesson prompt engineering taught you across this project?" → Good answer synthesis: hallucination and bias are not fixed properties of the model — they are heavily shaped by how the prompt is framed. Naive prompts default to the most dramatic/available answer; forcing named sources, explicit timeframes, and confidence scoring measurably improves reliability.

"What's a limitation of prompt engineering itself — where does it NOT help?" → It can't manufacture evidence that doesn't exist. If a source is genuinely unavailable or ambiguous, better prompting surfaces the uncertainty — it doesn't resolve it. (CS1's "unverified, not fabricated" category exists for exactly this reason.)

"Why did you use different prompting techniques for different sub-tasks instead of one universal 'best' prompt?" → Constraint-based zero-shot prompting suits factual extraction (risk = fabrication). CoT/ReAct suit synthesis tasks where a visible reasoning trace lets you check whether a conclusion is actually supported by evidence. Matching technique to failure mode, not using one hammer for everything.

"If you had to redo this with double the time, what would you change?" → Good honest answer: actually run the Phase 9 multi-LLM comparison live rather than leaving it templated, and expand the literature matrix

with full-text page-level detail rather than abstract-only synthesis (both documents note this themselves).

5. Viva Delivery Tips

Lead with the finding, not the process. If asked "what did you conclude," give the one-sentence takeaway first (e.g., "the trend is real but the numbers in circulation are inconsistently sourced — mostly stale figures, not fabrications"), *then* walk through phases if asked to elaborate.

Never claim a number you can't source on the spot. If pressed on a specific figure and you're not 100% sure, say "that's from [source] — I'd want to double check the exact figure before committing to it," rather than guessing.

Own the AI-usage transparently. These are prompt-engineering case studies — using AI heavily is the *point*, not something to hide. What matters is showing you understand *why* each technique was chosen and can critique its limits.

If asked to demonstrate live: have your Phase 8 Version 4 prompt (CS1) or a paper-verification search (CS2) ready to run in front of the examiner if you have a device. This is the single best way to preempt "did you actually do this yourself?"

Know your own document's weak points before they're pointed out. Better to say "one limitation of my own work here is..." than to be caught defending a gap you didn't notice.